Stacking galaxies reveals a universal baryon fountain

Averaging many simulated galaxy ecosystems exposes coherent gas inflows and outflows hidden by individual diversity.

arXiv:2609.21374 — Stacked baryon cycles in cosmological hydrodynamic simulations

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Background

Galaxies grow by cycling gas inward from their surroundings and expelling it through feedback, forming fountain-like flows. Individual systems are too diverse and dynamic to reveal the underlying pattern directly.

> **Observational hints of baryon cycles exist, but no clean, quantitative picture of the average flow geometry has emerged.**

What this paper is selling

The authors stack gas velocity and property fields across ensembles of simulated galaxies, suppressing scatter to reveal strikingly regular fountain patterns. A simple parameterized model then reproduces the stacked flow.

Key results to remember

- Stacking yields smooth, coherent inflow-outflow cones consistent with observed gas kinematics.
- Fountain geometry persists across stellar mass and redshift, with outflow cones widening at earlier times.
- A compact parameterized model matches the stacked velocity field, linking simulations to observations.

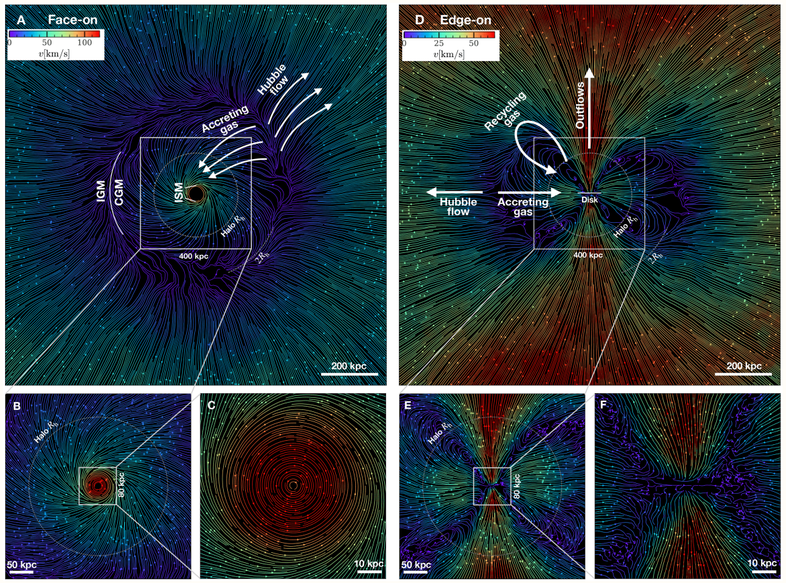


Figure 1 | Stacked streamlines show a coherent fountain: gas flowing in along the disk plane and out in bipolar cones.

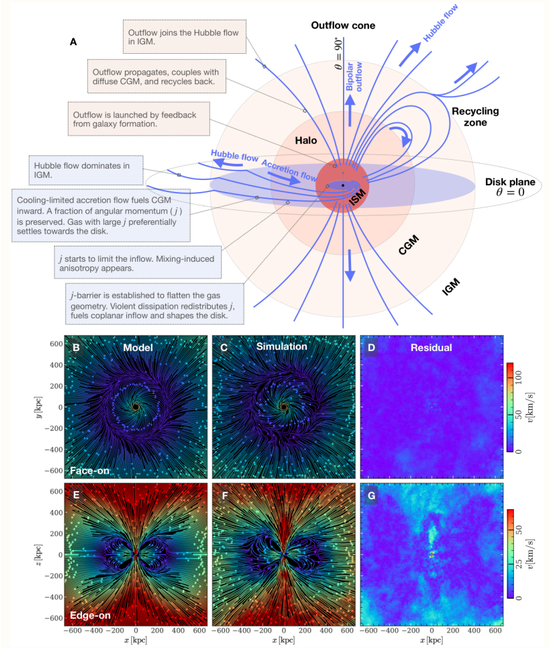


Figure 4 | A simple parameterized fountain model reproduces the stacked velocity field, with only small residuals.